WORKED SOLUTION

STEP I 2009, Question 7

Pure • Integration by Parts

> **STEP I 2009, Question 7**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/09-s1-q7

The key idea

The opening integral is the classic "integrate by parts twice and the integral comes back" trick: after two steps the original integral reappears on the right-hand side, and we solve for it like an unknown in an equation. Once we have a formula for $\int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x$ for *every* integer m, the later parts become bookkeeping: products of sines and cosines are turned into sums of single cosines, each of which we can integrate at once.

Solution

The general integral

Write $I_m = \int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x$, and let $E = \mathrm{e}^{2\pi} - 1$ for short. Integrating by parts with $u = \cos mx$ and $\mathrm{d}v = \mathrm{e}^x \,\mathrm{d}x$,

$$I_m = \Big[\mathrm{e}^x \cos mx\Big]_0^{2\pi} + m \int_0^{2\pi} \mathrm{e}^x \sin mx \,\mathrm{d}x.$$

Since m is an integer, $\cos 2m\pi = 1$, so the bracket is $\mathrm{e}^{2\pi} - 1 = E$. Integrating by parts again, now with $u = \sin mx$,

$$\int_0^{2\pi} \mathrm{e}^x \sin mx \,\mathrm{d}x = \Big[\mathrm{e}^x \sin mx\Big]_0^{2\pi} - m \int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x = 0 - mI_m,$$

because $\sin 2m\pi = \sin 0 = 0$, again using that m is an integer. Putting the two together,

$$I_m = E - m^2 I_m, \qquad \text{so} \qquad (m^2+1) I_m = E.$$

> **Ask yourself** We have gone round in a circle and got I_m back. Is that a dead end?

No: that is exactly what we want. The equation $I_m = E - m^2 I_m$ is a linear equation for

the unknown number I_m, and since $m^2+1 \neq 0$ we can divide by it:

$$\int_0^{2\pi} \mathrm{e}^x \cos mx \,\mathrm{d}x = \frac{1}{m^2+1}(\mathrm{e}^{2\pi}-1),$$

as required. (The case $m=0$ is included: it just says $\int_0^{2\pi} \mathrm{e}^x \,\mathrm{d}x = E$.) Notice that the integer condition was used to make the boundary terms simple; for non-integer m the formula is false.

Part (i)

Expanding with the compound-angle formulae, the $\sin A \sin B$ terms cancel:

$$\cos(A+B)+\cos(A-B) = 2\cos A \cos B.$$

With $A = 6x$ and $B = x$ this gives $\cos x \cos 6x = \frac{1}{2}(\cos 7x + \cos 5x)$. Hence, by the general result with $m = 7$ and $m = 5$,

$$\begin{aligned}\int_0^{2\pi} \mathrm{e}^x \cos x \cos 6x \,\mathrm{d}x &= \tfrac{1}{2}(I_7 + I_5) = \tfrac{1}{2}\Big(\frac{1}{50}+\frac{1}{26}\Big)E \\ &= \tfrac{1}{2}\cdot\frac{26+50}{1300}E = \frac{76}{2600}E = \frac{19}{650}(\mathrm{e}^{2\pi}-1),\end{aligned}$$

which is the required result.

Part (ii)

We need two product-to-sum steps. Subtracting rather than adding the compound-angle formulae gives $\cos(A-B)-\cos(A+B) = 2\sin A \sin B$, so with $A = 4x$, $B = 2x$,

$$\sin 2x \sin 4x = \tfrac{1}{2}(\cos 2x - \cos 6x).$$

Multiplying by $\cos x$ and using part (i)'s identity on each product,

$$\begin{aligned}\sin 2x \sin 4x \cos x &= \tfrac{1}{2}\cos 2x \cos x - \tfrac{1}{2}\cos 6x \cos x \\ &= \tfrac{1}{4}(\cos 3x + \cos x) - \tfrac{1}{4}(\cos 7x + \cos 5x).\end{aligned}$$

Now each term is covered by the general result, with $m = 3, 1, 7, 5$:

$$\begin{aligned}\int_0^{2\pi} \mathrm{e}^x \sin 2x \sin 4x \cos x \,\mathrm{d}x &= \tfrac{1}{4}(I_3 + I_1 - I_7 - I_5) \\ &= \tfrac{1}{4}\Big(\frac{1}{10}+\frac{1}{2}-\frac{1}{50}-\frac{1}{26}\Big)E \\ &= \tfrac{1}{4}\cdot\frac{65+325-13-25}{650}E = \tfrac{1}{4}\cdot\frac{352}{650}E.\end{aligned}$$

So

$$\int_0^{2\pi} \mathrm{e}^x \sin 2x \sin 4x \cos x \,\mathrm{d}x = \frac{44}{325}(\mathrm{e}^{2\pi}-1).$$

Checks

- Differentiating $\dfrac{e^x(\cos mx + m\sin mx)}{m^2+1}$ gives $e^x\cos mx$, so this is an antiderivative; evaluating it at 2π and 0 reproduces the general result.
- Put $x = 0$ in the part (ii) identity: the left side is 0 and the right side is $\frac{1}{4}(1+1-1-1) = 0$. Good.
- Numerically (Simpson's rule), the part (i) integral is $15.6236\ldots$ and the part (ii) integral is $72.3619\ldots$, matching $\frac{19}{650}E$ and $\frac{44}{325}E$ with $E = e^{2\pi} - 1 \approx 534.49$.

Common slips

- Sign errors in the second integration by parts: the derivative of $\sin mx$ is $+m\cos mx$, which with the minus sign from the parts formula gives $-m^2 I_m$.
- Forgetting the factor $\frac{1}{2}$ in each product-to-sum step, so that part (ii) picks up $\frac{1}{2}$ instead of $\frac{1}{4}$.

Beyond STEP

A slicker route is to write $\cos mx = \operatorname{Re} e^{imx}$, so that $I_m = \operatorname{Re}\int_0^{2\pi} e^{(1+im)x}\,dx = \operatorname{Re}\dfrac{E}{1+im} = \dfrac{E}{1+m^2}$, with no integration by parts at all. Breaking a function into its components $\cos mx$, as we did in the later parts, is the start of Fourier series, a central tool in first-year analysis and differential equations. Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction, and Needham, *Visual Complex Analysis*, shows why the exponential viewpoint works so well.

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